

BlinkIT Grocery Sales Analytics — Project Design Document

1. Document Control

Item	Details
Project	BlinkIT Grocery Sales Analytics
Dataset Records	8,523
Dataset Attributes	12
Unique Outlets	10
Item Categories	16

2. Executive Summary

The **BlinkIT Grocery Sales Analytics** project is a business intelligence solution designed to analyze grocery sales performance across products, outlets, outlet formats, outlet sizes, geographic tiers, fat-content categories, and outlet establishment years.

The solution converts a flat grocery-sales dataset into an interactive Power BI dashboard that enables users to monitor overall sales, compare outlet performance, analyze product categories, evaluate outlet characteristics, and interactively filter the results.

The supplied source contains **8,523 records and 12 attributes**, with **10 unique outlets and 16 item categories**.

3. Business Problem

Retail and grocery businesses generate large volumes of product and outlet data. Without a consolidated analytical view, it is difficult to answer:

1. What is the overall sales performance?
2. Which outlet formats contribute the most sales?
3. Which location tiers generate the highest sales?
4. How does outlet size relate to sales?
5. Which product categories are the strongest contributors?
6. How does sales performance vary by fat-content classification?
7. How does outlet establishment year relate to sales?
8. Which product/outlet combinations require additional investigation?

4. Project Objectives

Primary Objectives

- Build a professional Power BI dashboard for grocery sales analysis.
- Establish a consistent set of business KPIs.
- Provide interactive filtering and multidimensional exploration.
- Present sales performance from both outlet and product perspectives.
- Improve decision support for retail management.

Portfolio Objectives

- Demonstrate data preparation and analytical modeling.
- Demonstrate Power BI visualization and DAX capability.
- Provide professional project and solution documentation.
- Create a reusable structure for future retail analytics work.

5. Project Scope

In Scope

- Sales KPI reporting.
- Outlet-level analysis.
- Outlet location analysis.
- Outlet size analysis.
- Outlet type analysis.
- Product/item-type analysis.
- Item fat-content analysis.
- Outlet establishment-year analysis.
- Average rating analysis.
- Interactive slicers.
- Detailed analytical table/matrix.
- Data-quality considerations.
- Power BI semantic model and measures.

Out of Scope

- Customer-level behavior.
- Inventory optimization.
- Procurement planning.
- Profit/margin analysis.
- Customer segmentation.
- Real-time streaming.
- Forecasting.
- Machine-learning models.

The current dataset does not contain the fields required to reliably calculate profit, margin, customer lifetime value, or transaction-level time trends.

6. Stakeholders and Users

Stakeholder	Primary Need
Business / Retail Manager	Overall sales and outlet performance
Category Manager	Product-category contribution
Outlet Manager	Outlet-format and outlet-size comparison
Business Analyst	Multidimensional exploration
Data Analyst	Data quality and KPI validation
Portfolio Reviewer / Recruiter	Evidence of BI and analytical skills

7. Key Business Questions

- What are total sales and average sales?
- How many outlets are represented?
- What is the average rating?
- Which location tier contributes the most sales?
- Which outlet size contributes the most sales?
- Which outlet type is the strongest sales contributor?
- Which item categories generate the highest sales?

- How do Low Fat and Regular products compare?
- How does sales vary by establishment year?
- Which item/outlet combinations produce stronger results?

8. Key Performance Indicators

KPI	Definition
Total Sales	Sum of Sales
Total Outlets	Distinct count of Outlet Identifier
Average Sales	Average of Sales
Average Rating	Average of Rating

Recommended explicit measures:

Total Sales =

`SUM ( 'Table1'[Sales] )`

Total Outlets =

`DISTINCTCOUNT ( 'Table1'[Outlet Identifier] )`

Average Sales =

`AVERAGE ( 'Table1'[Sales] )`

Average Rating =

`AVERAGE ( 'Table1'[Rating] )`

9. Data Overview

Source Attributes

- Item Fat Content
- Item Identifier
- Item Type
- Outlet Establishment Year

- Outlet Identifier
- Outlet Location Type
- Outlet Size
- Outlet Type
- Item Visibility
- Item Weight
- Sales
- Rating

Data Characteristics

- Records: **8,523**
- Columns: **12**
- Unique outlets: **10**
- Item categories: **16**
- Missing Item Weight values: **1,463**
- Sales total: **1,201,681.49**
- Average row-level sales: **140.99**
- Average rating: **3.97**

10. Initial Analytical Findings

The supplied dataset indicates:

- **Supermarket Type1** is the largest outlet-format contributor, with approximately **787,550** in sales.
- **Tier 3** has the highest sales contribution among location tiers, at approximately **472,133**.
- **Medium-sized outlets** contribute the highest total sales among outlet sizes, at approximately **507,896**.
- **Fruits and Vegetables** is the highest-sales item category in the supplied data, followed by **Snack Foods**.
- Fat-content labels are inconsistent and should be standardized before production reporting.

These are descriptive observations, not causal conclusions.

11. Dashboard Design

The supplied PBIX uses a single analytical page containing KPI cards, slicers, comparative visuals, trend analysis, product analysis, and a detailed matrix.

Component	Purpose
Brand/Header	Dashboard identity
KPI Cards	Headline metrics
Outlet Location slicer	Filter by location tier
Outlet Size slicer	Filter by outlet size
Metric selector	Metric-driven analysis
Location comparison	Compare average sales by location
Outlet-size visual	Compare average sales by outlet size
Establishment-year visual	Analyze sales across establishment years
Fat-content visual	Compare sales by fat-content segment
Item-type visual	Compare average sales by product category
Detail matrix	Combine product and outlet dimensions with KPIs

12. Functional Requirements

FR-01 — KPI Reporting

Display Total Sales, Total Outlets, Average Sales, and Average Rating.

FR-02 — Outlet Filtering

Allow filtering by outlet location type and outlet size.

FR-03 — Metric Analysis

Support metric selection through the dashboard's metric-selection mechanism.

FR-04 — Product Analysis

Provide analysis by item type and fat-content classification.

FR-05 — Outlet Analysis

Provide analysis by outlet type, outlet size, location tier, and establishment year.

FR-06 — Detailed Analysis

Provide a matrix combining product and outlet dimensions with KPI measures.

13. Non-Functional Requirements

Requirement	Target
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Usability	Business-user friendly
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Performance	Responsive for supplied dataset
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Maintainability	Centralized measures and documented definitions
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Reproducibility	Source, PBIX, and documentation clearly separated
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Scalability	Model should support future refreshes
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Governance	KPI definitions and data-quality assumptions documented
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14. Assumptions

- Sales represents the primary sales-value metric.
- Outlet Identifier uniquely identifies an outlet.
- Rating is represented at item-record level in the source.
- Outlet Establishment Year is treated as an analytical year attribute.
- No profit or cost calculation is attempted because those fields are not present.

15. Data Quality Considerations

The source contains inconsistent item fat-content labels:

- Low Fat
- LF
- low fat
- Regular
- reg

Recommended standardization:

LF / Low Fat / low fat -> Low Fat

reg / Regular -> Regular

The source also contains **1,463 missing Item Weight values**. Since Item Weight is not required for the headline KPIs, it can remain nullable for this version.

19. Conclusion

The BlinkIT Grocery Sales Analytics project demonstrates an end-to-end BI workflow: data understanding, KPI definition, data-quality assessment, Power BI modeling, interactive visualization, and business-oriented insight generation.

The project provides a strong portfolio foundation that can evolve into a broader retail analytics platform when additional information such as transaction dates, costs, profit, inventory, promotions, and customer data becomes available.